

Tutorial on

Programming Novel AI Accelerators for Scientific Computing

SCA/HPC Asia
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Tutorial Agenda

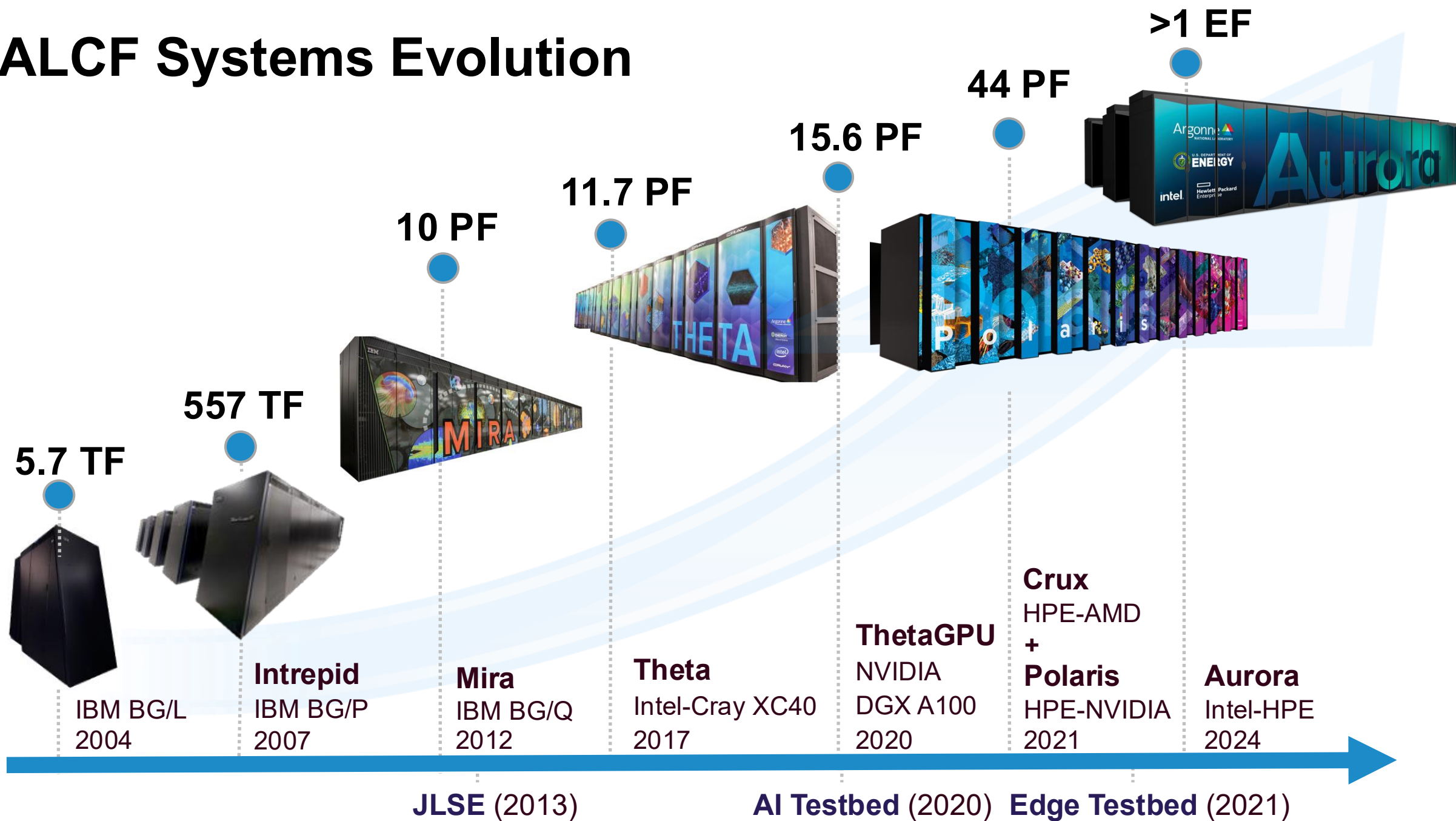
<https://github.com/argonne-lcf/Alaccelerators-SC25-tutorial/>

Agenda

Time (EST)	Topic/Speaker
9:30 AM - 9:45 AM	Welcome and Overview of the ALCF AI Testbed (Murali) Slides
9:45 AM - 10:45 AM	Cerebras (Leighton) Slides
10:45 AM - 11:15 AM	Coffee Break
11.15 AM - 12:00 PM	SambaNova (Tim) Slides
12:00 PM - 12:20 PM	Groq (Murali)
12.20 PM - 12:30 PM	Q&A and Conclusion (Murali)



ALCF Systems Evolution



ALCF AI Testbed

<https://www.alcf.anl.gov/alcf-ai-testbed>



Cerebras (CS-3)



SambaNova SN30/SN40L



Groq



Graphcore



Tenstorrent
Coming soon!!

- Infrastructure of next-generation machines with hardware accelerators customized for artificial intelligence (AI) applications.
- Provide a platform to evaluate usability and performance of machine learning based HPC applications running on these accelerators.
- The goal is to better understand how to integrate AI accelerators with ALCF's existing and upcoming supercomputers to accelerate science insights

ALCF AI Testbed

ALCF AI Testbed Systems are in production and available for allocations to the research community

Training

- Cerebras
- Sambanova SN30



SN-30 8 nodes of 8 RDUs



Cerebras CS-3 – 4 WSE

Inference

- SN40L – Metis
- Groq
- Cerebras
- Tenstorrent



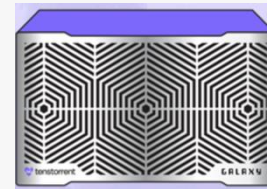
2 nodes of 16 SN40L RDUs



9 Groq nodes,
8 GroqChip/node (TSPs)



Cerebras CS-3 – 4 WSE



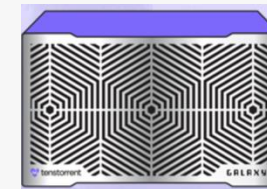
32 Wormhole GU

HPC

- Cerebras
- Tenstorrent



Cerebras CSL



32 Wormhole GU

Getting Started on ALCF AI Testbed

Available for Allocations

- Cerebras CS-3,
- SambaNova Datascale SN30,
- GroqRack
- Graphcore Bow Pod64
- Sambanova Inference – Metis SN40L (Available for all ALCF users via Inference Service Endpoints)

AI Testbed User Guide

Director's Discretionary (DD) awards

- Scaling code
- Preparing for future computing competition
- Scientific computing in support of strategic partnerships.

Allocation Request Form

<https://www.alcf.anl.gov/science/directors-discretionary-allocation-program>

NAIRR Pilot

Aims to connect U.S. researchers and educators to computational, data, and training resources needed to advance AI research and research that employs AI.

<https://nairrpilot.org/>

Machines > AI Testbed

<https://docs.alcf.anl.gov/ai-testbed/getting-started/>

ALCF AI Testbed



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The [ALCF AI Testbed](#) houses some of the most advanced AI accelerators for scientific research.

The goal of the testbed is to enable explorations into next-generation machine learning applications and workloads, enabling the ALCF and its user community to help define the role of AI accelerators in scientific computing and how to best integrate such technologies with supercomputing resources.

The AI accelerators complement the ALCF's current and next-generation supercomputers to provide a state-of-the-art computing environment that supports pioneering research at the intersection of AI, big data, and high performance computing (HPC).

The platforms are equipped with architectural features that support AI and data-centric workloads, making them well suited for research tasks involving the growing deluge of scientific data produced by powerful tools, such as supercomputers, light sources, telescopes, particle accelerators, and sensors. In addition, the testbed will allow researchers to explore novel workflows that combine AI methods with simulation and experimental science to accelerate the pace of discovery.

Useful Links

ALCF AI Testbed

- Overview: <https://www.alcf.anl.gov/alcf-ai-testbed>
- Guide: <https://docs.alcf.anl.gov/ai-testbed/getting-started/>
- Training:
 - <https://www.alcf.anl.gov/ai-testbed-training-workshops>
 - <https://github.com/argonne-lcf/Alaccelerators-SC25-tutorial/>
- Allocation Request: [Allocation Request Form](#)
- Support: support@alcf.anl.gov